

KARTHIK K

Software Developer

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PROFESSIONAL SUMMARY

Motivated and detail-oriented Software Developer and current MCA student with strong foundations in Java, Python, and JavaScript. Experienced in building and deploying full-stack web applications, RESTful APIs, and AI-integrated platforms. Comfortable working in collaborative, Agile team environments with version control (Git/GitHub) and containerised workflows (Docker). Eager to contribute clean, maintainable code while continuously learning and growing within a professional development team.

TECHNICAL SKILLS

- **Languages:** Java (Advanced), Python (Intermediate), JavaScript, HTML, CSS
- **Frameworks & Libraries:** React, FastAPI, Flask, TailwindCSS, Vite
- **Databases:** SQL, PostgreSQL, MongoDB, Firebase Firestore
- **Tools & Platforms:** Git, GitHub, Docker, Firebase Auth, APScheduler
- **Practices:** Agile / Scrum, Code Reviews, REST API Design, CI/CD (basic), Microservices, Documentation
- **AI & Integrations:** Google Gemini AI, NLP pipelines, Agentic AI concepts

WORK EXPERIENCE

AI Software Developer Intern | EnviroApps Inc (Remote) — Florida, USA

Dec 2025 – Present

- Contributed to developing the EAPHARMICS-ASEL ESTAR platform — a multi-module software system — by writing Python and FastAPI components for pharmaceutical data pipelines.
- Debugged and resolved backend issues across microservices, improving overall data processing reliability and reducing manual effort by an estimated 60%.
- Participated in Agile sprint cycles, code reviews, and asynchronous collaboration across international time zones, consistently delivering features on schedule.
- Wrote and maintained technical documentation for API endpoints, data schemas, and deployment processes, ensuring clear handoffs between sprint cycles.
- Assisted in containerising services with Docker for consistent, reproducible deployments across development and staging environments.

PROJECTS

Agentic AI for Audit Trail

Dec 2025 – Present

Stack: React, Python, FastAPI, PostgreSQL, Google Gemini AI, APScheduler, Docker

- Designed and developed a full-stack compliance microservice with FastAPI (Python) and PostgreSQL, containerised with Docker for reliable cross-environment deployment.
- Integrated background AI agents to autonomously monitor system logs and flag anomalies, cutting manual audit review effort by an estimated 70%.
- Maintained clear API and process documentation to support ongoing development and team onboarding.

Generative AI for Education

Aug 2025 – Nov 2025

Stack: React, Python, Flask, MongoDB, Google Gemini AI, NLP

- Built a full-stack personalised education platform integrating Google Gemini AI to dynamically generate study content, quizzes, and concept explanations.
- Implemented NLP-driven topic detection to auto-adjust content difficulty, reducing student research time by an estimated 50%.
- Designed RESTful API endpoints with Flask and MongoDB, handling concurrent sessions with low-latency responses.

ShareBox Marketplace

Sep 2025 – Nov 2025

Stack: React, Vite, TailwindCSS, Firebase Firestore & Auth

- Developed a hybrid e-commerce and donation marketplace using React and Firebase, with real-time Firestore database listeners for live inventory updates.
- Implemented Firebase Auth for secure user session management with sub-second listing refresh for all active users.
- Shipped a fully responsive, mobile-accessible UI with TailwindCSS and Vite, optimised for fast load times across device types.

EDUCATION

MCA (Pursuing) | Sona College of Technology, Salem
Current CGPA: 8.3

2024 – 2026

B.Sc. Computer Science | Vysya College, Salem
CGPA: 7.4

2019 – 2022

CERTIFICATIONS & TRAINING

- Java Full Stack Developer Training — QSpiders, Chennai (Nov 2022 – Dec 2023)
- Java — NPTEL (Elite + Silver) (Jan 2025 – Apr 2025)
- Python Programming — Infosys Springboard (Feb 2024)